

Midterm Conversion Prob 5

a) convert the num to Binary, Octal & hex

$$49.1875_{10}, 3.07421875_{10}, 0.2_{10}$$

Then convert them to Float representation

1) 49.1875_{10} To hex:

$$49_{10} \text{ to hex} \rightarrow 49 / 16 = 3 \text{ remainder } 1 \quad \uparrow$$

$$3 / 16 = 0 \text{ remainder } 3 \quad \uparrow$$

$$\text{so } 49_{10} = 31_{16}$$

$$0.1875_{10} \text{ to hex} \rightarrow .1875_{10} \times 16 = 3.0 \quad \downarrow$$

$$\text{so } 0.1875_{10} = 0.3_{16}$$

$$\text{so } 49.1875_{10} = 31.3_{16}$$

49.1875_{10} To Octal:

$$49 \text{ to Oct} \rightarrow 49 / 8 = 6 \text{ remainder } 1 \quad \uparrow$$

$$6 / 8 = 0 \text{ remainder } 6 \quad \uparrow$$

$$\text{so } 49_{10} = 61_8$$

$$0.1875_{10} \text{ to Oct} \rightarrow .1875_{10} \times 8 = 1.5 \rightarrow 1 \quad \downarrow$$

$$.5_{10} \times 8 = 4.0 \rightarrow 4 \quad \downarrow$$

$$\text{so } 0.1875_{10} = 0.14_8$$

$$\text{so } 49.1875_{10} = 61.14_8$$

49.1875_{10} To Binary:

$$49 \text{ to Binary} \rightarrow 49_{10} = 61_8 = 110\ 001_2$$

$$0.1875_{10} \text{ to Binary} \rightarrow .1875_{10} \times 2 = 0.375 \rightarrow 0 \quad \downarrow$$

$$.375_{10} \times 2 = 0.75 \rightarrow 0 \quad \downarrow$$

$$.75_{10} \times 2 = 1.5 \rightarrow 1 \quad \downarrow$$

$$.5_{10} \times 2 = 1.0 \rightarrow 1 \quad \downarrow$$

$$\text{so } 0.1875_{10} = .0011_2$$

$$\text{so } 49.1875_{10} = 110\ 001.0011_2$$

49.1875₁₀ in Float Representation:

$$49.1875_{10} = 110001.0011_2$$
$$= \underline{.11000100110} \times 2^6$$

$$= \underbrace{01100010}_6 \underbrace{0110}_2 \underbrace{0000}_6 \underbrace{0000}_0 \underbrace{0000}_0 \underbrace{00000110}_6_2$$

so $\boxed{49.1875_{10} = 62600006_{16}}$ in NASA Format

2.) $3.07421875_{10} \rightarrow \text{hex}$

$$3_{10} = 3_{16}$$

$$0.07421875_{10} \rightarrow .07421875_{10} \times 16 = 1.1875 \rightarrow 1$$
$$.1875_{10} \times 16 = 3.0 \rightarrow 3$$

$$\text{So } \boxed{3.07421875_{10} = 3.13_{16}}$$

$3.07421875_{10} \rightarrow \text{Octal}$

$$3_{10} = 3_8$$

$$0.07421875_{10} \rightarrow .07421875_{10} \times 8 = 0.59375 \rightarrow 0$$
$$.59375_{10} \times 8 = 4.75 \rightarrow 4$$
$$.75_{10} \times 8 = 6.0 \rightarrow 6$$

$$\text{So } \boxed{3.07421875_{10} = 3.46_8}$$

$3.07421875_{10} \rightarrow \text{Binary}$

$$3_{10} = 11_2$$

$$0.07421875_{10} \rightarrow .07421875_{10} \times 2 = 0.1484375 \rightarrow 0$$
$$.1484375 \times 2 = 0.296875 \rightarrow 0$$
$$.296875 \times 2 = 0.59375 \rightarrow 0$$
$$.59375 \times 2 = 1.1875 \rightarrow 1$$
$$.1875 \times 2 = 0.375 \rightarrow 0$$
$$.375 \times 2 = 0.75 \rightarrow 0$$
$$.75 \times 2 = 1.5 \rightarrow 1$$
$$.5 \times 2 = 1.0 \rightarrow 1$$

$$\text{So } \boxed{3.07421875_{10} = 11.00010011_2}$$

3.07421875 in Float Representation

$$3.07421875_{10} = 11.00010011_2$$
$$= 1.100010011 \times 2^2$$

$$= 01100010 \mid 01100000 \mid 00000000 \mid 00000010$$

6 2 6 0 0 0 0 2

so $\boxed{3.07421875_{10} = 62600002_{16}}$ in
Float Rep.

3. 0.2_{10} in hex

$$0.2_{10} \rightarrow 0.2 \times 16 = 3.2 \rightarrow 3$$

$$.2 \times 16 = 3.2 \rightarrow 3$$

$$\text{so } 0.2_{10} = 0.\underline{3}_{16}$$

0.2_{10} in Octal

$$0.2_{10} \rightarrow 0.2 \times 8 = 1.6 \rightarrow 1$$

$$.6 \times 8 = 4.8 \rightarrow 4$$

$$.8 \times 8 = 6.4 \rightarrow 6$$

$$.4 \times 8 = 3.2 \rightarrow 3$$

$$.2 \times 8$$

repeating $\leftarrow$

$$\text{so } 0.2_{10} = 0.\underline{1463}_8$$

0.2_{10} in Binary

$$0.2_{10} \rightarrow 0.2 \times 2 = 0.4 \rightarrow 0$$

$$.4 \times 2 = 0.8 \rightarrow 0$$

$$.8 \times 2 = 1.6 \rightarrow 1$$

$$.6 \times 2 = 1.2 \rightarrow 1$$

$$.2 \times 2$$

repeating $\leftarrow$

$$\text{so } 0.2_{10} = 0.\underline{0011}_2$$

0.2₁₀ in Float representation

$$0.2_{10} = 0.0011$$

$$0.110 \ 0110 \times 2^{-2}$$

↓ -2

$$\begin{array}{r} 11111111 \\ - \quad \quad 10 \\ \hline 11111101 \\ +1 \\ \hline 11111110 \end{array}$$

$$\begin{array}{cccccc} 0110 & 0110 & | & 0110 & 0110 & | & 0110 & 0110 & | & 11111110 \\ 6 & 6 & & 6 & 6 & & 6 & 6 & & F \ E \end{array}$$

$$\boxed{0.2_{10} = 666666FE} \text{ in Nasa Format}$$

b) -49.1875 in Dec negative Format

999.9999

-49.1875

950.8124

+0.0001

950.8125

$$\text{so } \boxed{-49.1875_{10} = 950.8125_{10}}$$

-49.1875 in hex negative Format

$$-49.1875_{10} = -31.3_{16}$$

FFF.F

-31.3

FCE.C

+0.1

FCE.D

$$\text{so } \boxed{-49.1875_{10} = \text{FCE.D}_{16}}$$

-49.1875 in Octal negative Format

$$-49.1875 = -61.14_8$$

777.77

-61.14

716.63

+0.01

716.64

$$\text{so } \boxed{-49.1875_{10} = 716.64_8}$$

-49.1875₁₀ in Binary negative format

$$-49.1875_{10} = -110001.0011_2$$

$$\begin{array}{r} 111111.1111 \\ -110001.0011 \\ \hline 100110.1100 \\ +.0001 \\ \hline 100110.1101 \end{array}$$

$$-49.1875_{10} = 100110.1101_2$$

-49.1875₁₀ in Float Representation

$$\begin{aligned} -49.1875_{10} &= 100110.1101_2 \\ &= 1.001101101 \times 2^6 \end{aligned}$$

$$= 1001101 / 10100000 / 00000000 / 00000110$$

9 0 A 0 0 0 0 6

$$-49.1875_{10} = 90A00006 \text{ in Hex Format}$$

-3.07421875₁₀ in Dec negative Format

99.99999999

- 3.07421875

96.92578124

+ .00000001

96.92578125

$$\text{so } \boxed{-3.07421875_{10} = 96.92578125_{10}}$$

-3.07421875₁₀ in hex negative Format

$$-3.07421875_{10} = -3.13_{16}$$

FF.FF

-3.13

FC.EC

+ .01

FC.E0

$$\text{so } -3.07421875_{10} = \boxed{\text{FC.E0}_{16}}$$

-3.07421875₁₀ in Octal negative Format

$$-3.07421875_{10} = -3.46_8$$

77.77

-3.46

74.31

+ .01

74.32

$$\text{so } -3.07421875_{10} = \boxed{74.32_8}$$

-3.07421875₁₀ in Binary Negative Form

$$-3.07421875_{10} = -11.00010011_2$$

$$\begin{array}{r} 111.1111111 \\ -11.00010011 \\ \hline 100.11101100 \\ +.00000001 \\ \hline 100.11101101 \end{array}$$

$$\boxed{\text{so } -3.07421875_{10} = 100.11101101_2}$$

-3.07421875₁₀ in Float Representation

$$\begin{aligned} -3.07421875_{10} &= 100.11101101_2 \\ &= 1.0011101101 \times 2^2 \end{aligned}$$

$$= 10011101 \mid 10100000 \mid 00000000 \mid 00000010$$

9 D A 0 0 0 0 2

$$\text{so } \boxed{-3.07421875_{10} = 9DA00002} \text{ in NASA Format}$$

-0.2₁₀ in Dec Negative Format

$$\begin{array}{r} 9.9 \\ - 0.2 \\ \hline 9.7 \\ + 0.1 \\ \hline 9.8 \end{array}$$

$$\text{so } \boxed{-0.2_{10} = 9.8_{10}}$$

-0.2₁₀ in Hex Negative Format

$$-0.2_{10} = -0.3_{16}$$

$$\begin{array}{r} F.F \\ - 0.3 \\ \hline F.C \\ + 0.1 \\ \hline F.D \end{array}$$

$$\text{so } \boxed{-0.2_{10} = F.D_{16}}$$

-0.2₁₀ in Octal Negative Format

$$-0.2_{10} = -0.1463_8$$

$$\begin{array}{r} 7.7777 \\ - 0.1463 \\ \hline 7.6314 \\ + 0.0001 \\ \hline 7.6315 \end{array}$$

$$\text{so } \boxed{-0.2_{10} = 7.6315_8}$$

-0.2_{10} in Binary negative format

$$-0.2_{10} = -0.0011_2$$

$$\begin{array}{r} 1.1111 \\ - 0.0011 \\ \hline 1.1100 \\ + 0.0001 \\ \hline 1.1101 \end{array}$$

$$\text{so } \boxed{-0.2_{10} = 1.1101_2}$$

-0.2_{10} in Float Representation

$$\begin{aligned} -0.2_{10} &= 1.1101_2 \\ &= 1.1101 \times 2^0 \end{aligned}$$

$$= \begin{array}{cccc|cccc|cccc|cccc} 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline E & E & E & E & E & E & E & E & E & E & E & E & E & E & E & E & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$$\text{so } \boxed{-0.2_{10} = E E E E E E 00} \text{ in NASA format}$$

$$c) 69999902 =$$

$$0110 \ 1001 \ | \ 1001 \ 1001 \ | \ 1001 \ 1001 \ | \ \underbrace{0000 \ 0010}_2$$

$$= 0.1101001 \times 2^2$$

$$= 11.01001100$$

$$= 3.4 \underline{C}_{16}$$

$$= 3 + \frac{4}{16} + \frac{C}{16^2} + \frac{C}{16^3} + \frac{C}{16^4} + \dots$$

$$= 3 + \frac{4}{16} + \left(\frac{C}{16} \times \sum_{n=1}^{\infty} \frac{1}{16^n} \right)$$

$$= 3 + \frac{4}{16} + \left(\frac{12}{16} \times \frac{1}{15} \right)$$

$$= \frac{720 + 60 + 12}{240} = \frac{792}{240} = \underline{\underline{3.3_{10}}}$$

69999903

$$= 01101001 \mid 10011001 \mid 10011001 \mid 00000011$$

$$= 0.1101001 \times 2^3$$

$$= 0110.1001_2$$

$$= 6.\underline{9}_{16}$$

$$= 6 + \frac{9}{16} + \frac{9}{16^2} + \frac{9}{16^3} + \dots$$

$$= 6 + \left(\frac{9}{16} \times \sum_{n=1}^{\infty} \frac{1}{16^n} \right) = 6 + \left(\frac{9}{16} \times \frac{1}{15} \right) = \frac{1449}{240}$$

$$\boxed{= 6.0375_{10}}$$

$$966667FF$$

$$= 1001\ 0110\ |0110\ 0110\ |0110\ 0111\ |1111\ 1111$$

$$= 1.\underline{0010}\underline{1100}\underline{1100}\underline{1100}\underline{1100}\underline{1110} \times 2^{-127}$$

$$= -0.00000000\ 0000\ 0000\ 0000\ 0000\ 0000\ 1 \times 2^{-128}$$

$$= 1.\underline{0010}\underline{1100}\underline{1100}\underline{1100}\underline{1100}\underline{1101} \times 2^{-127}$$

$$= 1.\underline{1111}\underline{1111}\underline{1111}\underline{1111}\underline{1111}\underline{1111} \times 2^{-127}$$

$$= -0.\underline{0010}\underline{1100}\underline{1100}\underline{1100}\underline{1100}\underline{1100} \times 2^{-127}$$

$$= -0.5587701667 \times 2^{-127}$$

$$= -0.0010\ 1100 \times 2^{-127}$$

$$= -0.00010\ 110 \times 2^{-128}$$

$$= -0.16 \times 16^{-32}$$

$$= -\frac{1}{16} + \left(\frac{6}{16} \times \left(\sum_{n=1}^{\infty} \frac{1}{16^n} \right) \right) = \frac{1}{16} + \left(\frac{6}{16} \times \frac{1}{15} \right) =$$

$$= -0.0001041667 \times 10^{-32}$$

$$= \underline{-0.1041667 \times 10^{-35}}$$